

Erdős Problem #291

EP-291

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $n \geq 1$ and define L_n to be the least common multiple of $\{1, \dots, n\}$ and a_n by $\sum_{1 \leq k \leq n} \frac{1}{k} = \frac{a_n}{L_n}$. Is it true that $(a_n, L_n) = 1$ and $(a_n, L_n) > 1$ both occur for infinitely many n ?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>